Split-Controlled Completion Lift Note

Internal technical note

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Abstract

This note packages a narrow completion-side statement on the accepted welded branch. Assuming an explicit accepted welded split package in the band-locked, co-centered regime, it proves that the inherited ledger algebra is QND, the abstract accepted-record payload on the inherited ledger is preserved exactly, and recoherence is localized in branchwise completion data B. The split package itself is imported structural data; the note does not derive it from microscopic UOM dynamics.

Contents

1 Scope **1**

2 Accepted Welded Split Package **3**

3 Exact Ledger-Side Consequences **4**

4 Bounded Split-Controlled Packaging Theorem **5**

5 Split-Controlled Scalar Descendant Lift **6**

6 Structural Limits **15**

1 Scope

Statement proved here. Fix an accepted welded branch in the band-locked, co-centered regime of proposition 1.3. This note proves a conditional completion-side statement: given an explicit accepted welded split package, the inherited ledger algebra is QND, the abstract accepted-record payload on the inherited ledger is preserved exactly, and recoherence is localized in branchwise completion data B.

Imported source basis. The inherited accepted-record algebra and its canonical record projection are imported from the UOM main manuscript (Definition 4.3, Lemma 4.1, Proposition 4.2, Remark 4.2). The structural $A|B$ split picture is imported from the summability note: the inherited-ledger/completion-sector identification from section 7.5, the controlled split form from (26), and the newborn coherence target from (27). The accepted welded branch narrowing comes from proposition 1.3, the accepted-record payload and witness-seam language come from definitions 4.26 and 4.27 and remark 4.28, and the ledger/completion compatibility statement comes from remark 3.24.

Boundary of the note. The only genuinely structural input left below is a branchwise completion realization that refines the inherited accepted-record projectors into completion fibers and puts the split into controlled form on those fibers. Everything else is imported directly from the current UOM source basis.

Definition 1.1 (Inherited accepted-record algebra at the split tick)**.** Fix one accepted tick k on an accepted welded branch in the band-locked, co-centered regime of proposition 1.3. Let $\mathcal{H}_{\mathrm{acc}}$ denote the accepted band-limited receiver space at that tick. By Definition 4.3, Lemma 4.1, Proposition 4.2, and Remark 4.2, $\mathcal{H}_{\mathrm{acc}}$ carries a commutative inherited accepted-record algebra with orthogonal record projectors $\{\Pi_i\} \subset \mathrm{End}(\mathcal{H}_{\mathrm{acc}})$ and canonical record projection

$$\mathsf{P}_{\mathrm{rec}}(X) := \sum_i \Pi_i X \Pi_i, \qquad X \in \mathrm{End}(\mathcal{H}_{\mathrm{acc}}).$$

We refer to this commutative algebra as the inherited accepted-record algebra at tick k.

Definition 1.2 (Branchwise newborn pointer projections)**.** Once a branchwise completion fiber is read as part of the next aeon's source, let $\Pi_{\Delta,i}$ denote the canonical pointer/record projection on that branchwise newborn realization in the sense of Definition 4.3. The coherence target imported from (27) is measured branchwise with these projections.

Assumption 1.3 (Controlled completion branch decomposition)**.** Fix one accepted tick k on an accepted welded branch in the band-locked, co-centered regime of proposition 1.3, and let the inherited accepted-record algebra be as in definition 1.1. Assume the structural $A|B$ split picture of section 7.5 in the following explicit form. There exist an abstract ledger space $\mathcal{H}_A$ with orthonormal basis $\{e_i\}$, rank-one ledger projectors $E_i^{(A)} := |e_i\rangle\langle e_i|$, branchwise completion fibers $\{\mathcal{H}_i^{(B)}\}$, and a unitary identification

$$W : \mathcal{H}_{\mathrm{acc}} \xrightarrow{\sim} \bigoplus_i \big(\mathbb{C}e_i \otimes \mathcal{H}_i^{(B)}\big)$$

such that:

1. the inherited accepted-record algebra is realized branchwise as

$$\mathrm{alg}\{\Pi_i\} = \mathrm{alg}\{W^\dagger (E_i^{(A)} \otimes \mathbf{1}_{\mathcal{H}_i^{(B)}}) W\},$$

   so in particular $W\Pi_i W^\dagger = E_i^{(A)} \otimes \mathbf{1}_{\mathcal{H}_i^{(B)}}$ on the i-th summand;

2. the split is implemented by a branchwise controlled unitary

$$W U_{\mathrm{split}} W^\dagger = \bigoplus_i \big(E_i^{(A)} \otimes U_i^{(B)}\big),$$

   with each $U_i^{(B)}$ unitary on $\mathcal{H}_i^{(B)}$, as the branchwise direct-sum form of (26).

Remark 1.4 (What remains assumed)*.* The inherited accepted-record algebra and its canonical projection are source-backed by the UOM main manuscript. What remains assumed in assumption 1.3 is only a direct-sum realization

$$\mathcal{H}_{\mathrm{acc}} \cong \bigoplus_i \big(\mathbb{C}e_i \otimes \mathcal{H}_i^{(B)}\big)$$

that realizes the inherited ledger projectors as branch labels and places the split in branchwise controlled form.

2 Accepted Welded Split Package

Definition 2.1 (Explicit accepted welded split package)**.** Under definitions 1.1 and 1.2 and assumption 1.3, an *explicit accepted welded split package* is a choice of:

1. an abstract inherited ledger register $\mathcal{H}_A$ with basis $\{e_i\}$ indexed by the inherited accepted-record projectors and rank-one projectors
$$E_i^{(A)} := |e_i\rangle\langle e_i|;$$
2. a family of branchwise completion fibers
$$\{\mathcal{H}_i^{(B)}\};$$
3. a unitary branch decomposition
$$W : \mathcal{H}_{\mathrm{acc}} \xrightarrow{\sim} \bigoplus_i \big(\mathbb{C}e_i \otimes \mathcal{H}_i^{(B)}\big),$$
with $W\Pi_i W^\dagger = E_i^{(A)} \otimes \mathbf{1}_{\mathcal{H}_i^{(B)}}$ on the i-th summand;
4. the commutative inherited ledger algebra
$$\mathcal{A}_{\mathrm{rec}} := \mathrm{alg}\{\Pi_i\} \subset \mathrm{End}(\mathcal{H}_{\mathrm{acc}});$$
5. a split unitary of branchwise controlled form
$$W U_{\mathrm{split}} W^\dagger = \bigoplus_i \big(E_i^{(A)} \otimes U_i^{(B)}\big),$$
with each $U_i^{(B)}$ unitary on $\mathcal{H}_i^{(B)}$;
6. the abstract accepted-record compression
$$\mathsf{P}^{\mathrm{acc}}(\rho) := \sum_i \mathrm{Tr}(\Pi_i \rho)\, E_i^{(A)}, \qquad \rho \in \mathcal{T}_1(\mathcal{H}_{\mathrm{acc}});$$
7. a family of branchwise newborn pointer projections
$$\{\Pi_{\Delta,i}\}.$$

We refer to the family $\{\mathcal{H}_i^{(B)}\}$ as the branchwise completion data B. An observable $L \in \mathcal{A}_{\mathrm{rec}}$ will be called *accepted-record visible*.

Definition 2.2 (Ledger-classical split input)**.** A state ρ on $\mathcal{H}_{\mathrm{acc}}$ is a *ledger-classical split input* if, under an explicit accepted welded split package, it has the branchwise classical-control form
$$W\rho W^\dagger = \sum_i p_i\, E_i^{(A)} \otimes \sigma_i^{(B)}, \qquad p_i \geq 0, \quad \sum_i p_i = 1,$$
where $\sigma_i^{(B)}$ is a state on $\mathcal{H}_i^{(B)}$ and each term is understood on the i-th direct summand of $\bigoplus_i(\mathbb{C}e_i \otimes \mathcal{H}_i^{(B)})$. Equivalently, the abstract inherited ledger register A acts as classical control for the branchwise completion preparation.

Definition 2.3 (Recoherence localized in branchwise completion data B)**.** Let $\rho' := U_{\text{split}}\rho U_{\text{split}}^{\dagger}$ under an explicit accepted welded split package, and write

$$\rho_A := \mathsf{P}^{\text{acc}}(\rho), \qquad \rho'_A := \mathsf{P}^{\text{acc}}(\rho').$$

For each i with $p'_i := \mathrm{Tr}(\Pi_i\rho') > 0$, let $\rho_i^{(B)\prime}$ be the unique state on $\mathcal{H}_i^{(B)}$ determined by

$$E_i^{(A)} W\rho' W^{\dagger} E_i^{(A)} = p'_i\, E_i^{(A)} \otimes \rho_i^{(B)\prime}.$$

Recoherence is said to be *localized in branchwise completion data B* if:

1. the abstract inherited ledger state is unchanged, i.e.

$$\rho'_A = \rho_A;$$

2. for at least one accepted branch i with $p'_i > 0$, the branchwise completion state satisfies the newborn coherence target

$$\left\| \rho_i^{(B)\prime} - \Pi_{\Delta,i}(\rho_i^{(B)\prime}) \right\|_1 > 0.$$

   This is the branchwise form of the coherence criterion imported from (27).

3 Exact Ledger-Side Consequences

Lemma 3.1 (Ledger observables are QND invariants)**.** *Let $(\mathcal{H}_A, \mathcal{A}_{\text{rec}}, U_{\text{split}}, \mathsf{P}^{\text{acc}})$ be an explicit accepted welded split package. Then every accepted-record-visible observable $L \in \mathcal{A}_{\text{rec}}$ satisfies*

$$U_{\text{split}}^{\dagger} L U_{\text{split}} = L.$$

In particular,

$$[U_{\text{split}}, \Pi_i] = 0 \qquad \textit{for every } i.$$

Proof. Under the explicit branch decomposition of definition 2.1,

$$W U_{\text{split}} W^{\dagger} = \bigoplus_i \big(E_i^{(A)} \otimes U_i^{(B)}\big),$$

so $W U_{\text{split}} W^{\dagger}$ commutes with each $E_i^{(A)} \otimes \mathbf{1}_{\mathcal{H}_i^{(B)}}$. Transporting this identity back to $\mathcal{H}_{\text{acc}}$ yields

$$[U_{\text{split}}, \Pi_i] = 0 \qquad \forall i.$$

Since $\mathcal{A}_{\text{rec}} = \mathrm{alg}\{\Pi_i\}$, the same identity implies $U_{\text{split}}^{\dagger} L U_{\text{split}} = L$ for every $L \in \mathcal{A}_{\text{rec}}$. □

Proposition 3.2 (Exact preservation of the accepted-record part)**.** *Let $(\mathcal{H}_A, \mathcal{A}_{\text{rec}}, U_{\text{split}}, \mathsf{P}^{\text{acc}})$ be an explicit accepted welded split package, and let*

$$\rho' := U_{\text{split}}\rho U_{\text{split}}^{\dagger}$$

for an arbitrary state ρ on $\mathcal{H}_{\text{acc}}$. Then the abstract accepted-record payload is preserved exactly:

$$\mathsf{P}^{\text{acc}}(\rho') = \mathsf{P}^{\text{acc}}(\rho).$$

Equivalently, every inherited ledger branch weight is preserved:

$$\mathrm{Tr}(\Pi_i\rho') = \mathrm{Tr}(\Pi_i\rho) \qquad \forall i.$$

Proof. By lemma 3.1, U_{split} commutes with every Π_i. Therefore

$$\mathrm{Tr}(\Pi_i \rho') = \mathrm{Tr}(\Pi_i U_{\text{split}} \rho U_{\text{split}}^\dagger) = \mathrm{Tr}(U_{\text{split}}^\dagger \Pi_i U_{\text{split}} \rho) = \mathrm{Tr}(\Pi_i \rho).$$

Multiplying each preserved coefficient by $E_i^{(A)}$ and summing over i yields the claimed identity for P^{acc}. □

Corollary 3.3 (Exact ledger preservation on classical-control input)**.** *Let* $(\mathcal{H}_A, \mathcal{A}_{\text{rec}}, U_{\text{split}}, \mathsf{P}^{\text{acc}})$ *be an explicit accepted welded split package and let* ρ *be a ledger-classical split input. Write*

$$\rho' := U_{\text{split}} \rho U_{\text{split}}^\dagger.$$

Then

$$W \rho' W^\dagger = \sum_i p_i \, E_i^{(A)} \otimes U_i^{(B)} \sigma_i^{(B)} (U_i^{(B)})^\dagger,$$

so in particular

$$\rho'_A = \rho_A = \sum_i p_i \, E_i^{(A)},$$

where $\rho_A := \mathsf{P}^{\text{acc}}(\rho)$ *and* $\rho'_A := \mathsf{P}^{\text{acc}}(\rho')$.

Proof. Substitute the classical-control form of $W \rho W^\dagger$ into the branchwise controlled expression for $W U_{\text{split}} W^\dagger$. All cross terms vanish because $E_i^{(A)} E_j^{(A)} = 0$ for $i \neq j$. The displayed formula follows immediately, and the identity for $\rho'_A = \mathsf{P}^{\text{acc}}(\rho')$ is then exactly proposition 3.2. □

4 Bounded Split-Controlled Packaging Theorem

Theorem 4.1 (Ledger-side consequences of explicit split data)**.** *Let an explicit accepted welded split package be given on an accepted welded branch, let* ρ *be the input state, and write*

$$\rho' := U_{\text{split}} \rho U_{\text{split}}^\dagger.$$

Assume that:

1. *the input state is a ledger-classical split input;*
2. *the accepted-record payload on the inherited receiver ledger is read through the accepted-record compression* P^{acc}*, equivalently through the accepted-record-visible algebra* $\mathcal{A}_{\text{rec}}$*;*
3. *the post-split branchwise completion data satisfy the newborn coherence target, i.e. there exists an accepted branch* i *with* $p'_i = \mathrm{Tr}(\Pi_i \rho') > 0$ *and branchwise completion state* $\rho_i^{(B)\prime}$ *as in definition 2.3 such that*

$$\left\| \rho_i^{(B)\prime} - \Pi_{\Delta,i}(\rho_i^{(B)\prime}) \right\|_1 > 0.$$

   as in (27)*.*

Then the following ledger-side consequences hold:

1. *the inherited ledger projectors are QND under the split;*
2. *the abstract accepted-record payload on* A *is preserved exactly, and for ledger-classical input one also has* $\rho'_A = \rho_A$*;*

3. *recoherence is localized in branchwise completion data B.*

Together with the imported split package itself, these consequences furnish the bounded split-controlled completion package used in this note.

Proof. Item (1) is lemma 3.1. Item (2) follows from proposition 3.2; under the present assumptions, the accepted-record payload is exactly the part read by $\mathsf{P}^{\mathrm{acc}}$. The stronger identity $\rho'_A = \rho_A$ for ledger-classical input is exactly corollary 3.3. Item (3) follows from corollary 3.3 together with the assumed branchwise completion coherence target, which is exactly the definition of recoherence localized in branchwise completion data B. □

Corollary 4.2 (Compatibility with the weak transported interface)**.** *Under the hypotheses of theorem 4.1, the split acts trivially on every accepted-record-visible observable in $\mathcal{A}_{\mathrm{rec}}$. Consequently, the completion-side recoherence considered here is compatible with the weak transported interface of theorem 3.20: the inherited ledger data remain unchanged, while the nontrivial completion-side effect is carried by branchwise completion data B, in accord with remark 3.24.*

Proof. By theorem 4.1, the split acts trivially on every $L \in \mathcal{A}_{\mathrm{rec}}$ and preserves the accepted-record part on A. The only recoherence admitted by the theorem is the branchwise B-localized recoherence of definition 2.3. This is exactly the separation between inherited ledger data and completion-side oddness recorded in remark 3.24, and it does not alter the ledger-visible data read by the weak transported interface. □

5 Split-Controlled Scalar Descendant Lift

Definition 5.1 (Split response scalar line)**.** Fix an accepted tick k and the next accepted receiver step $k+1$. Under the explicit accepted welded split package of definition 2.1, a *split response scalar line* at Λ_{k+1} is a one-dimensional real line

$$\mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}} := \mathbb{R}\, s_{\Lambda_{k+1}}.$$

It is understood as the scalar line extracted from the branchwise completion recoherence data at tick k and read at the next receiver step $k+1$.

Definition 5.2 (Split-controlled scalar descendant lift datum)**.** Fix an accepted tick k and the next accepted receiver step $k+1$. A *split-controlled scalar descendant lift datum* consists of:

(a) the delayed selected line

$$\widehat{L}^{\mathrm{sel,delay}}_k = \mathbb{R}\, \widehat{\ell}^{\mathrm{delay}}_k$$

of definition 2.2;

(b) a split response scalar line

$$\mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}} = \mathbb{R}\, s_{\Lambda_{k+1}}$$

as in definition 5.1;

(c) a source-compatible descendant representative

$$J^{\mathrm{split}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^{-}_{\Lambda_{k+1}};$$

(d) a linear map

$$\mathfrak{L}^{\mathrm{split}}_{k\to k+1} : \widehat{L}^{\mathrm{sel,delay}}_k \longrightarrow \mathfrak{D}^-_{\Lambda_{k+1}} := E^-_{\Lambda_{k+1}} / \mathcal{A}^-_{\Lambda_{k+1}}$$

such that

$$\mathfrak{L}^{\mathrm{split}}_{k\to k+1}\big(\widehat{\ell}^{\mathrm{delay}}_k\big) = q_{\Lambda_{k+1}}\big(J^{\mathrm{split}}_{\Lambda_{k+1}}\big),$$

where $q_{\Lambda_{k+1}}$ is the quotient map of definition 4.11;

(e) compatibility of the current Main-visible reduced welded image with the delayed scalar:

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{split}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

Definition 5.3 (Canonical split-to-descendant scalar bridge)**.** Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the delayed selected line $\widehat{L}^{\mathrm{sel,delay}}_k$ of definition 2.2 and a split response scalar line $\mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}}$ of definition 5.1. A *canonical split-to-descendant scalar bridge* consists of:

(a) a linear scalar-transport map

$$\mathfrak{T}^{\mathrm{split}}_{k\to k+1} : \widehat{L}^{\mathrm{sel,delay}}_k \longrightarrow \mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}}$$

such that the image

$$s^{\mathrm{split,can}}_{\Lambda_{k+1}} := \mathfrak{T}^{\mathrm{split}}_{k\to k+1}\big(\widehat{\ell}^{\mathrm{delay}}_k\big)$$

generates the split response scalar line:

$$\mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}} = \mathbb{R}\, s^{\mathrm{split,can}}_{\Lambda_{k+1}};$$

(b) a linear descendant lift map

$$\mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}} : \mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}} \longrightarrow E^-_{\Lambda_{k+1}}$$

such that the lifted scalar generator

$$J^{\mathrm{split,can}}_{\Lambda_{k+1}} := \mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}}\big(s^{\mathrm{split,can}}_{\Lambda_{k+1}}\big)$$

lies in the odd descendant class:

$$J^{\mathrm{split,can}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}};$$

(c) compatibility of the lifted scalar generator with the current Main-visible reduced welded image:

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

Thus the only transport-level input not already canonical in the note stack is an explicit bridge that identifies the canonical delayed selected generator with a canonical split scalar generator and lifts that scalar generator into the odd descendant class. The later bridge-construction theorem of this section shows that, once the source-compatible unit comparison and canonical-representative packages are imported, this bridge is produced canonically from the B-localized recoherence output itself.

Proposition 5.4 (A canonical scalar bridge induces canonical scalar datum)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume a canonical split-to-descendant scalar bridge in the sense of definition 5.3. Define the descended bridge map by*

$$\mathfrak{L}^{\mathrm{split,can}}_{k\to k+1} := q_{\Lambda_{k+1}} \circ \mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}} \circ \mathfrak{T}^{\mathrm{split}}_{k\to k+1} \;:\; \widehat{L}^{\mathrm{sel,delay}}_k \longrightarrow \mathfrak{D}^-_{\Lambda_{k+1}},$$

where $q_{\Lambda_{k+1}}$ is the quotient map of definition 4.11. Set

$$\overline{\ell}^{\mathrm{Main,can}}_{\Lambda_{k+1}} := q_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big)$$

and

$$\overline{L}^{\mathrm{Main,can}}_{\Lambda_{k+1}} := \mathbb{R}\,\overline{\ell}^{\mathrm{Main,can}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}}.$$

Then:

1. *the data*

$$\widehat{L}^{\mathrm{sel,delay}}_k, \quad \mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}}, \quad J^{\mathrm{split,can}}_{\Lambda_{k+1}}, \quad \mathfrak{L}^{\mathrm{split,can}}_{k\to k+1}$$

   together with $\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}} P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}$ form a split-controlled scalar descendant lift datum in the sense of definition 5.2;

2. *the element $J^{\mathrm{split,can}}_{\Lambda_{k+1}}$ is canonical relative to the bridge, because it is determined by the canonical delayed selected generator $\widehat{\ell}^{\mathrm{delay}}_k$ together with $\mathfrak{T}^{\mathrm{split}}_{k\to k+1}$ and $\mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}}$;*

3. *the scalar line $\overline{L}^{\mathrm{Main,can}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}}$ is canonical relative to the bridge.*

Proof. Because $\widehat{L}^{\mathrm{sel,delay}}_k = \mathbb{R}\,\widehat{\ell}^{\mathrm{delay}}_k$ by definition 2.2, the scalar-transport map $\mathfrak{T}^{\mathrm{split}}_{k\to k+1}$ determines the generator $s^{\mathrm{split,can}}_{\Lambda_{k+1}}$ uniquely. Applying $\mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}}$ then determines $J^{\mathrm{split,can}}_{\Lambda_{k+1}}$ uniquely, and by definition of a canonical split-to-descendant scalar bridge this element lies in $[\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}$ and satisfies the reduced welded compatibility clause of definition 5.2.

The map $\mathfrak{L}^{\mathrm{split,can}}_{k\to k+1}$ is linear by construction, and

$$\mathfrak{L}^{\mathrm{split,can}}_{k\to k+1}\big(\widehat{\ell}^{\mathrm{delay}}_k\big) = q_{\Lambda_{k+1}}\big(\mathfrak{J}^{\mathrm{split}}_{\Lambda_{k+1}}\big(\mathfrak{T}^{\mathrm{split}}_{k\to k+1}\big(\widehat{\ell}^{\mathrm{delay}}_k\big)\big)\big) = q_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big),$$

which is exactly the remaining clause of definition 5.2. So item (1) follows.

Items (2) and (3) are immediate from the displayed definitions: no auxiliary basis choice on $\mathfrak{S}^{\mathrm{split}}_{\Lambda_{k+1}}$ or $\mathfrak{D}^-_{\Lambda_{k+1}}$ remains after the bridge is fixed. □

Theorem 5.5 (Canonical split scalar carrier from B-localized recoherence)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume:*

1. *the hypotheses and conclusions of theorem 4.1;*

2. *the delayed selected-line data of definitions 2.2 and 3.1.*

Set

$$\mathfrak{S}^{\mathrm{split,can}}_{\Lambda_{k+1}} := \widehat{L}^{\mathrm{sel,delay}}_k = \mathbb{R}\,\widehat{\ell}^{\mathrm{delay}}_k, \qquad s^{\mathrm{split,can}}_{\Lambda_{k+1}} := \widehat{\ell}^{\mathrm{delay}}_k,$$

and define

$$\mathfrak{T}^{\mathrm{split}}_{k\to k+1} := \mathrm{id}_{\widehat{L}^{\mathrm{sel,delay}}_k} : \widehat{L}^{\mathrm{sel,delay}}_k \longrightarrow \mathfrak{S}^{\mathrm{split,can}}_{\Lambda_{k+1}}.$$

Then:

1. $\mathfrak{S}^{\mathrm{split,can}}_{\Lambda_{k+1}}$ *is a split response scalar line in the sense of definition 5.1;*

2. *the generator* $s^{\mathrm{split,can}}_{\Lambda_{k+1}}$ *is canonically determined by the unit delayed selected generator exported by the delayed selected-line package;*

3. $\mathfrak{T}^{\mathrm{split}}_{k\to k+1}$ *is a canonical linear isomorphism from the delayed selected line to the split scalar carrier.*

Proof. By definition 2.2, $\widehat{L}^{\mathrm{sel,delay}}_k = \mathbb{R}\,\widehat{\ell}^{\mathrm{delay}}_k$ is already a one-dimensional real line with canonical generator. By theorem 4.1, the only post-split nontrivial effect certified in the present note is the recoherence localized in branchwise completion data B, while the inherited A-side ledger is preserved exactly. So the honest scalar carrier extracted from that B-localized recoherence is precisely the canonical delayed selected carrier already isolated at the next receiver step. Declaring $\mathfrak{S}^{\mathrm{split,can}}_{\Lambda_{k+1}}$ to be that line itself therefore furnishes a split response scalar line in the sense of definition 5.1, and the identity map is a canonical linear isomorphism. □

Definition 5.6 (Source-compatible unit descendant realization)**.** Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the unit seam source $\tau^{\mathrm{unit}}_{\Lambda_{k+1}}$ is representable on the characterization layer in the sense of theorem 4.34, with representable source generator $F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}} \in E^-_{\Lambda_{k+1}}$. A representative

$$J^{\mathrm{unit}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}$$

is called a *source-compatible unit descendant realization* if

$$J^{\mathrm{unit}}_{\Lambda_{k+1}} - F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}} \in \mathcal{A}^-_{\Lambda_{k+1}}.$$

Equivalently, $J^{\mathrm{unit}}_{\Lambda_{k+1}}$ and $F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}$ define the same point of the ambiguity quotient $\mathfrak{D}^-_{\Lambda_{k+1}} = E^-_{\Lambda_{k+1}}/\mathcal{A}^-_{\Lambda_{k+1}}$.

Theorem 5.7 (Canonical source-side quotient bridge from the unit welded odd source)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume:*

1. *the hypotheses of theorem 5.5;*

2. *the unit welded odd source satisfies the specialized representability and scalar identity of* **??** *at* Λ_{k+1}*;*

3. *the descended dominant functional on* $\mathfrak{D}^-_{\Lambda_{k+1}}$ *exists, for example by the generator-wise vanishing package of proposition 4.15.*

Define the source-side quotient bridge point by

$$\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}} := q_{\Lambda_{k+1}}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}\big) \in \mathfrak{D}^-_{\Lambda_{k+1}}$$

and define

$$\overline{L}^{\mathrm{split,src}}_{\Lambda_{k+1}} := \mathbb{R}\,\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}} \subset \mathfrak{D}^-_{\Lambda_{k+1}}, \qquad \mathfrak{L}^{\mathrm{split,src}}_{k\to k+1} := \big(a\,\widehat{\ell}^{\mathrm{delay}}_k\big) \longmapsto a\,\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}.$$

Then:

1. *the quotient point* $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$ *is canonically determined by the representable unit welded odd source;*

2. $\mathfrak{L}^{\mathrm{split,src}}_{k\to k+1}$ *is a canonical linear map from the delayed selected line to the descendant ambiguity quotient, sending* $\widehat{\ell}^{\mathrm{delay}}_{k}$ *to* $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$*;*

3. *the descended dominant scalar of that quotient point is the unit seam coefficient:*

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}\big) = \rho^{\mathrm{dom}}_{\Lambda_{k+1}}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}.$$

Hence the B-localized recoherence package determines a canonical source-side split-to-descendant quotient bridge datum before any descendant representative is chosen.

Proof. **??** already identifies the representable generator $F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}$ canonically and proves

$$\rho^{\mathrm{dom}}_{\Lambda_{k+1}}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}.$$

Because the descended dominant functional on $\mathfrak{D}^{-}_{\Lambda_{k+1}}$ exists, its universal factorization identity gives

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(q_{\Lambda_{k+1}}(J)\big) = \rho^{\mathrm{dom}}_{\Lambda_{k+1}}(J) \qquad \forall J \in E^{-}_{\Lambda_{k+1}}.$$

Applying this to $J = F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}$ proves item (3). Items (1) and (2) are then immediate from the displayed definitions and theorem 5.5. □

Proposition 5.8 (Ambiguity-subordination criterion for the source-side bridge point)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses of theorem 5.7. Let*

$$J^{\mathrm{unit}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^{-}_{\Lambda_{k+1}}.$$

Then the following are equivalent:

(a)

$$J^{\mathrm{unit}}_{\Lambda_{k+1}} - F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}} \in \mathcal{A}^{-}_{\Lambda_{k+1}};$$

(b)

$$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit}}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}.$$

If these equivalent statements hold, then

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(q_{\Lambda_{k+1}}(J^{\mathrm{unit}}_{\Lambda_{k+1}})\big) = c_{\Lambda_{k+1}} \qquad \textit{and} \qquad \mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{unit}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

Proof. Statement (a) is equivalent to

$$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit}}_{\Lambda_{k+1}}\big) = q_{\Lambda_{k+1}}\big(F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}}\big),$$

which is exactly statement (b) by the definition of $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$. So the two formulations are equivalent.

If they hold, then

$$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit}}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}.$$

Applying item (3) of theorem 5.7 gives

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(q_{\Lambda_{k+1}}(J^{\mathrm{unit}}_{\Lambda_{k+1}})\big) = c_{\Lambda_{k+1}}.$$

Because $J^{\mathrm{unit}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}]$, corollary 4.13 identifies the left-hand side with $\rho^{\mathrm{dom}}_{\Lambda_{k+1}}(J^{\mathrm{unit}}_{\Lambda_{k+1}})$. Finally, proposition 4.46 gives the displayed reduced welded image identity. □

Theorem 5.9 (Canonical unit realization from one subordinate realization)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume:*

1. *the hypotheses of proposition 5.8;*
2. *there exists a source-compatible unit descendant realization*
$$J^{\mathrm{unit},0}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}$$
satisfying the equivalent conditions of proposition 5.8;
3. *the imported finite-cutoff canonical representative package at Λ_{k+1}: there exists a Hilbertizing bilinear form $\mathsf{B}_{\Lambda_{k+1}}$ on $E^-_{\Lambda_{k+1}}$ and a unique representative*
$$\mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}$$
minimizing $\mathsf{B}_{\Lambda_{k+1}}(J,J)$, equivalently satisfying
$$\mathsf{B}_{\Lambda_{k+1}}\big(\mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}}, A\big) = 0 \qquad \forall\, A \in \mathcal{A}^-_{\Lambda_{k+1}}.$$

Set
$$J^{\mathrm{unit,can}}_{\Lambda_{k+1}} := \mathcal{J}^{\mathrm{can}}_{\Lambda_{k+1}}.$$
Then:

1. $J^{\mathrm{unit,can}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^-_{\Lambda_{k+1}}$;
2. $$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit,can}}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}};$$
3. $$J^{\mathrm{unit,can}}_{\Lambda_{k+1}} - F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}} \in \mathcal{A}^-_{\Lambda_{k+1}};$$
4. $J^{\mathrm{unit,can}}_{\Lambda_{k+1}}$ *is canonical relative to $\mathsf{B}_{\Lambda_{k+1}}$ and the source-side bridge point $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$.*

Hence one subordinate unit realization is enough to promote the fixed quotient bridge point to a canonical source-compatible unit descendant realization.

Proof. Item (1) is immediate from the definition of $J^{\mathrm{unit,can}}_{\Lambda_{k+1}}$. Because $J^{\mathrm{unit,can}}_{\Lambda_{k+1}}$ and $J^{\mathrm{unit},0}_{\Lambda_{k+1}}$ both lie in the same affine class $[\Xi^{(1)}_{\Lambda_{k+1}}]$, their difference belongs to $\mathcal{A}^-_{\Lambda_{k+1}}$. Since $J^{\mathrm{unit},0}_{\Lambda_{k+1}}$ satisfies the equivalent conditions of proposition 5.8, we have
$$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit},0}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$$
and
$$J^{\mathrm{unit},0}_{\Lambda_{k+1}} - F_{\tau^{\mathrm{unit}}_{\Lambda_{k+1}},\Lambda_{k+1}} \in \mathcal{A}^-_{\Lambda_{k+1}}.$$
Adding the ambiguity difference $J^{\mathrm{unit,can}}_{\Lambda_{k+1}} - J^{\mathrm{unit},0}_{\Lambda_{k+1}} \in \mathcal{A}^-_{\Lambda_{k+1}}$ to the second display proves item (3). Applying the quotient map to the same ambiguity difference proves item (2). Finally, item (4) is exactly the uniqueness clause in the imported finite-cutoff canonical-representative package. □

Theorem 5.10 (Bridge closure from one ambiguity-subordinated unit realization)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses of theorem 5.9. Then the source-side quotient bridge point $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}}$ is exactly the descendant-class quotient point $q_{\Lambda_{k+1}}(J^{\mathrm{unit,can}}_{\Lambda_{k+1}})$, and the canonical map $\mathfrak{L}^{\mathrm{split,src}}_{k\to k+1}$ is the desired split-to-descendant bridge datum in $\mathfrak{D}^{-}_{\Lambda_{k+1}}$.*

Proof. This is exactly item (2) of theorem 5.9, together with the definition of $\mathfrak{L}^{\mathrm{split,src}}_{k\to k+1}$ from theorem 5.7. □

Corollary 5.11 (Subordinate unit realizations are unique modulo ambiguity)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses of proposition 5.8. Let*

$$J^{\mathrm{unit},1}_{\Lambda_{k+1}}, \quad J^{\mathrm{unit},2}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}] \cap E^{-}_{\Lambda_{k+1}}$$

each satisfy the equivalent conditions of proposition 5.8. Then:

1. $$J^{\mathrm{unit},2}_{\Lambda_{k+1}} - J^{\mathrm{unit},1}_{\Lambda_{k+1}} \in \mathcal{A}^{-}_{\Lambda_{k+1}};$$

2. $$q_{\Lambda_{k+1}}\big(J^{\mathrm{unit},1}_{\Lambda_{k+1}}\big) = q_{\Lambda_{k+1}}\big(J^{\mathrm{unit},2}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}};$$

3. $$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(q_{\Lambda_{k+1}}(J^{\mathrm{unit},1}_{\Lambda_{k+1}})\big) = \overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(q_{\Lambda_{k+1}}(J^{\mathrm{unit},2}_{\Lambda_{k+1}})\big) = c_{\Lambda_{k+1}};$$

4. $$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{unit},1}_{\Lambda_{k+1}}\big) = \mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{unit},2}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

In particular, source-compatible unit realizations subordinate to the canonical source-side bridge point are unique modulo $\mathcal{A}^{-}_{\Lambda_{k+1}}$, and every datum read only through that quotient point together with the reduced welded image is representative-independent on that class.

Proof. Because both $J^{\mathrm{unit},1}_{\Lambda_{k+1}}$ and $J^{\mathrm{unit},2}_{\Lambda_{k+1}}$ belong to $[\Xi^{(1)}_{\Lambda_{k+1}}]$, their difference lies in $\mathcal{A}^{-}_{\Lambda_{k+1}}$, which proves item (1). Item (2) is exactly the quotient-point clause of proposition 5.8 applied to each representative. Items (3) and (4) are then the final conclusions of proposition 5.8, again applied to each representative. The concluding sentence is immediate from those displayed identities. □

Remark 5.12 (Exact-family accepted-boundary invariance)*.* The theorem-backed corollary above controls the quotient point and the reduced welded image. On the current exact split-controlled surrogate family, the separate audit packet `Main/Mathematica/Reports/Lane1BridgeClosureAudit.md` also certifies a stronger finite-dimensional statement: any two admissible positive-calibration realizations subordinate to the same source-side bridge point have the same accepted welded boundary image. That accepted-boundary invariance is therefore presently exact-family support rather than theorem-backed text.

Corollary 5.13 (Canonical representative reconstruction after bridge closure)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses of theorem 5.10. Set*

$$J^{\mathrm{split,can}}_{\Lambda_{k+1}} := J^{\mathrm{unit,can}}_{\Lambda_{k+1}}$$

and define

$$\mathfrak{I}^{\mathrm{split}}_{\Lambda_{k+1}} : \mathfrak{S}^{\mathrm{split,can}}_{\Lambda_{k+1}} \longrightarrow E^{-}_{\Lambda_{k+1}}, \qquad \mathfrak{I}^{\mathrm{split}}_{\Lambda_{k+1}}\big(a\, s^{\mathrm{split,can}}_{\Lambda_{k+1}}\big) := a\, J^{\mathrm{split,can}}_{\Lambda_{k+1}}.$$

Then:

1. $$q_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big) = \overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}};$$

2. $$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}\big(J^{\mathrm{split,can}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}};$$

3. *the pair* $\mathfrak{T}^{\mathrm{split}}_{k\to k+1}$ *and* $\mathfrak{I}^{\mathrm{split}}_{\Lambda_{k+1}}$ *defines a canonical split-to-descendant scalar bridge in the sense of definition 5.3.*

Proof. Item (1) is exactly theorem 5.10. Item (2) is the reduced welded-image conclusion of proposition 5.8, applied to the canonical unit realization supplied by theorem 5.9.

By theorem 5.5, the generator $s^{\mathrm{split,can}}_{\Lambda_{k+1}}$ is exactly $\widehat{\ell}^{\mathrm{delay}}_k$ and $\mathfrak{T}^{\mathrm{split}}_{k\to k+1}$ is the identity on the delayed selected line. So every clause of definition 5.3 is satisfied, proving item (3). □

Corollary 5.14 (B-localized recoherence plus one subordinate unit realization produces canonical scalar datum)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume the hypotheses of corollary 5.13. Then proposition 5.4 applies with the bridge reconstructed in corollary 5.13. Consequently the split-controlled delayed selected-line data determine a canonical scalar datum and a canonical quotient subspace*

$$\overline{L}^{\mathrm{Main,can}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

Proof. Apply proposition 5.4 to the canonical bridge furnished by corollary 5.13. □

Theorem 5.15 (Split-controlled scalar descendant lift)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume:*

1. *the hypotheses and conclusions of theorem 4.1;*
2. *the delayed selected-line package of definitions 2.2 and 3.1 and theorem 4.2;*
3. *a split-controlled scalar descendant lift datum of definition 5.2.*

Set

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} := \mathfrak{L}^{\mathrm{split}}_{k\to k+1}\big(\widehat{\ell}^{\mathrm{delay}}_k\big) = q_{\Lambda_{k+1}}\big(J^{\mathrm{split}}_{\Lambda_{k+1}}\big)$$

and

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} := \mathbb{R}\,\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

Then:

1. *the map*
$$\mathfrak{L}^{\mathrm{split}}_{k\to k+1} : \widehat{L}^{\mathrm{sel,delay}}_k \to \mathfrak{D}^{-}_{\Lambda_{k+1}}$$
is well defined on the delayed selected line;

2. *the image generator is nonzero:*
$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0;$$

3. *the image generator is dominant-active:*
$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}\big(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}\big) = c_{\Lambda_{k+1}} \neq 0;$$

4. $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}$ *is faithful on* $\widehat{L}^{\mathrm{sel,delay}}_k$*;*

5. *the current Main-visible reduced welded image of the lifted generator is nonzero:*

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) \neq 0.$$

Proof. By theorem 4.2, the delayed realization map has trivial kernel on $\widehat{L}_k^{\mathrm{sel,delay}}$. So the delayed selected line is already faithful on its own delayed carrier.

Because $\widehat{L}_k^{\mathrm{sel,delay}} = \mathbb{R}\,\widehat{\ell}_k^{\mathrm{delay}}$ by definition 2.2, any linear map out of that line is determined uniquely by the image of the generator. The datum of definition 5.2 therefore makes $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}$ well defined, proving item (1).

By the compatibility clause in definition 5.2,

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}}\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

Here $c_{\Lambda_{k+1}} \neq 0$ by the delayed selected-line package and $P^{\mathrm{LoG}}_{-,\Lambda_{k+1}} \neq 0$ by corollary 1.10. Hence the displayed reduced welded image is nonzero, proving item (5).

Now suppose for contradiction that

$$\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} = q_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = 0.$$

Then $J^{\mathrm{split}}_{\Lambda_{k+1}} \in \mathcal{A}^-_{\Lambda_{k+1}}$, so proposition 4.46 gives

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = 0,$$

contradicting item (5) already proved. Therefore $\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$, which is item (2).

By proposition 4.46, the reduced welded image is exactly the dominant-response scalar times the canonical LoG projector:

$$\mathsf{W}^{\mathrm{red}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = \rho^{\mathrm{dom}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}})\, P^{\mathrm{LoG}}_{-,\Lambda_{k+1}}.$$

Comparing with the compatibility clause from definition 5.2 shows

$$\rho^{\mathrm{dom}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}}.$$

Because $J^{\mathrm{split}}_{\Lambda_{k+1}} \in [\Xi^{(1)}_{\Lambda_{k+1}}]$, corollary 4.13 yields

$$\overline{\rho}^{\mathrm{dom}}_{\Lambda_{k+1}}(\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}}) = \rho^{\mathrm{dom}}_{\Lambda_{k+1}}(J^{\mathrm{split}}_{\Lambda_{k+1}}) = c_{\Lambda_{k+1}} \neq 0,$$

which proves item (3).

Finally, the domain $\widehat{L}_k^{\mathrm{sel,delay}}$ is one-dimensional, and its generator has nonzero image $\overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} \neq 0$. Therefore the kernel of $\mathfrak{L}^{\mathrm{split}}_{k\to k+1}$ is trivial, proving item (4). □

Corollary 5.16 (Canonical split-controlled scalar descendant lift)**.** *Fix an accepted tick k and the next accepted receiver step $k+1$. Assume:*

1. *the hypotheses of corollary 5.14;*

2. *the delayed selected-line package of definitions 2.2 and 3.1 and theorem 4.2.*

Then the canonical datum of corollary 5.14 satisfies the hypotheses of theorem 5.15. Consequently the conclusions of that theorem hold with

$$J^{\mathrm{split}}_{\Lambda_{k+1}} := J^{\mathrm{split,can}}_{\Lambda_{k+1}}, \qquad \overline{\ell}^{\mathrm{Main}}_{\Lambda_{k+1}} := \overline{\ell}^{\mathrm{Main,can}}_{\Lambda_{k+1}},$$

and

$$\overline{L}^{\mathrm{Main}}_{\Lambda_{k+1}} = \overline{L}^{\mathrm{Main,can}}_{\Lambda_{k+1}} \subset \mathfrak{D}^{-}_{\Lambda_{k+1}}.$$

In particular, once the canonical split-to-descendant scalar bridge is supplied, the scalar datum of the split-controlled delayed selected-line package is canonical and determines a canonical nonzero dominant-active scalar line in the ambiguity quotient once the delayed selected-line package is in force.

Proof. By corollary 5.14, the canonical bridge built from the B-localized recoherence output induces a split-controlled scalar descendant lift datum. The remaining assumptions of theorem 5.15 are already included in corollary 5.14, except for the delayed selected-line package, which is assumed here. So the conclusions of theorem 5.15 apply verbatim to the canonical datum. □

Remark 5.17 (Boundary for canonical scalar data). Theorem theorem 5.7 determines the canonical source-side bridge point $\overline{\ell}^{\mathrm{split,src}}_{\Lambda_{k+1}} \in \mathfrak{D}^{-}_{\Lambda_{k+1}}$ before any descendant representative is chosen. If there exists a source-compatible unit descendant realization subordinate to that bridge point, then Theorem theorem 5.9 and Corollary corollary 5.11 canonicalize that realization and reconstruct the canonical split-to-descendant scalar bridge.

What remains assumed is the existence of one such subordinate source-compatible unit descendant realization. Nor does the present note derive the package nondegeneracy clause $c_{\Lambda_{k+1}} \neq 0$; that clause is still imported from the upstream seam-overlap/transport lane.

Remark 5.18 (Scope of the scalar bridge). theorem 5.15 reaches only the ambiguity quotient $\mathfrak{D}^{-}_{\Lambda_{k+1}}$ and the current Main-visible reduced welded image. It does *not* claim any automatic further reentry into a stronger selected descendant quotient, physical quotient, or mod-null package. That later step remains conditional.

6 Structural Limits

Remark 6.1 (Conditional character). The proofs in this note begin once the controlled completion branch decomposition of assumption 1.3 is granted. They show that the inherited ledger projectors are QND, the abstract accepted-record payload on A is preserved exactly, and any recoherence certified here is confined to branchwise completion data B. They do not derive the direct-sum realization $\mathcal{H}_{\mathrm{acc}} \cong \bigoplus_i (\mathbb{C}e_i \otimes \mathcal{H}_i^{(B)})$ or its branchwise controlled realization from microscopic UOM dynamics.

Remark 6.2 (Residual source-internal factorization problem). To remove the explicit split-data premise from theorem 4.1, one would need a source-internal statement on the accepted welded branch asserting that the compiled welded control is determined entirely by the accepted-record sectors of the inherited ledger, so that the controlling branch labels and branchwise completion actions are fixed by accepted-record data with no additional completion-side choice. That factorization problem is separate from the ledger-side consequences proved here. In particular, the present theorem should not be read as identifying the full completion datum with the current accepted-record sector.

Remark 6.3 (Gaussian-independent scope). Nothing in this note proves Gaussian dilation existence, symplectic uniqueness, or completion Hamiltonian reconstruction. Those realization questions are not used in the proofs of lemma 3.1, proposition 3.2, corollary 3.3, and theorem 4.1.